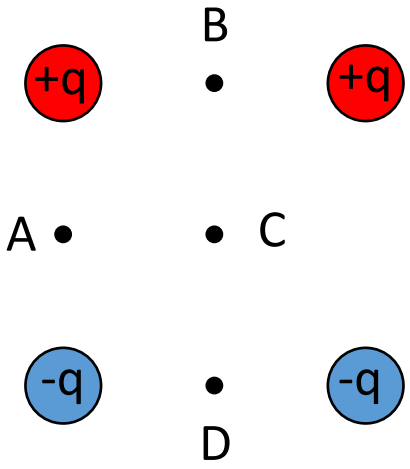
Exercise 1.



Four point charges having values of +/- q are arranged in a square as shown in the figure. The points A, B, D are placed at the midpoints between the nearest point charges, whereas point C is in the center of the square. How do the electrostatic potentials V_A-V_D in the different points relate to each other?

A) $V_A > V_B > V_C > V_D$

B) $V_D > V_C > V_B > V_A$

C) $V_B > V_A > V_C > V_D$

D) $V_B = V_D > V_A > V_C$

E) $V_B = V_D > V_A = V_C$

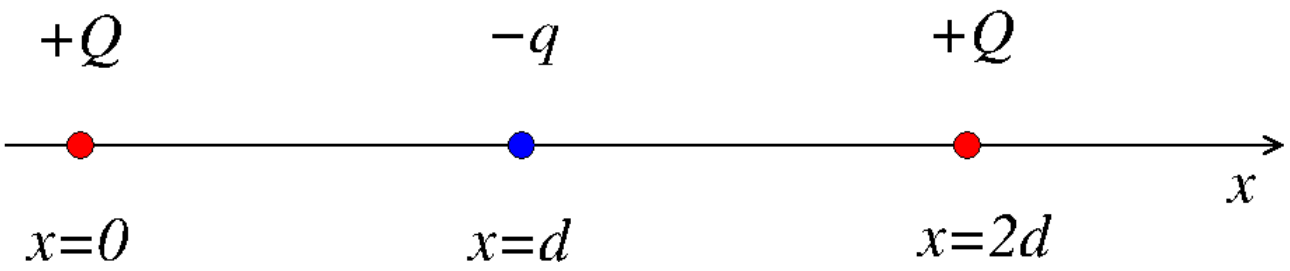
F) $V_B > V_A = V_C > V_D$

G) $V_D > V_A = V_C > V_B$

H) $V_A = V_B = V_C = V_D$

I) Don’t know

Exercise 2.



Three point charges lie on a line with neares-neighbour separation *d* as shown in the figure. The relationship between the magnitudes of the charges *Q* and *q* is such that the charges are in static equilibrium. When calculating electrostatic potential, we assume that the potential goes to 0 at infinity. Which of the statements below are correct?

A) $Q=2q$

B) $Q=4q$

C) The electrostatic potential has a zero point on the *x*-axis at $x=d/2$, i.e. midway between particle 1 and 2.

D) The electrostatic potential has a zero point on the *x*-axis at $x=4d/5$.

E) The electrostatic potential has exactly two zero points on the *x*-axis.

F) A test charge placed at a zero point of the electrostatic potential will not be affected by an electrostatic force.

G) Don't know.

Exercise 3.

The below table gives mass density, ρ_m molar mass, M, and number of conduction electrons per atom, Z, for the metals Ag, Cu, Al (silver, copper, aluminium).

Metal	Z	ρ_m (kg/m^3)	M (kg/mole)
Ag	1	$1.05 \cdot 10^4$	0.108
Cu	1	$8.80 \cdot 10^3$	0.0635
Al	3	$2.70 \cdot 10^3$	0.0270

a) Derive an expression for the drift velocity of the charge carriers, v_d in a cable having a circular cross section, as a function of the current I, and the cable (conductor) diameter d. Plot the drift velocity as a function of d for I=10.0 A, in the three materials, e.g. for d between 0.1 and 100 mm.

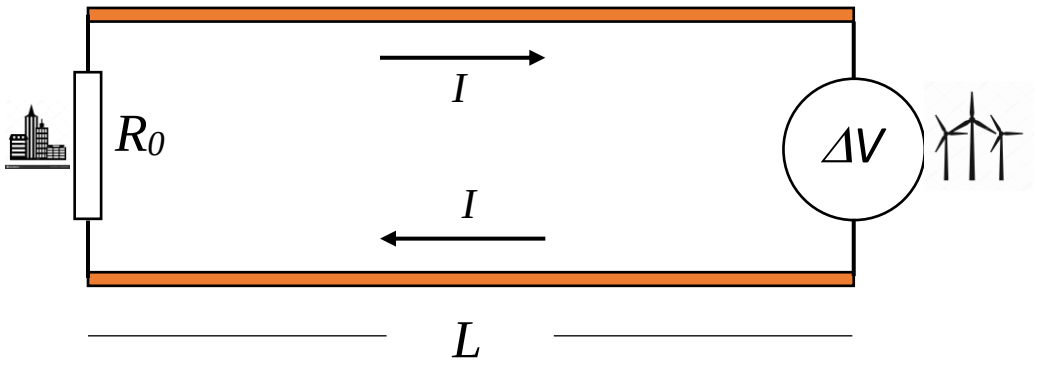
b) Determine the resistance per length as a function of d in the three materials. Data for resistivity can be found in the textbook, Table 9.1.

c) Determine the mass per meter for a cable with resistance 0.1 Ω/km for the three materials.

Exercise 4.

Power transmission over long distances is becoming increasingly important in future energy systems, because it can help stabilizing electrical grids with a large amount of fluctuating power sources. 'High Voltage Direct Current' (HVDC), finds increasing use for such applications.

In this exercise we analyze a highly simplified model of such a system. Two conductors of length L connects the power source in the right part of the figure (e.g. a power plant, or a wind farm) with end users represented by the resistive load R_0. The voltage difference between the conductors at the power source is denoted ΔV. We assume the power source delivers a power of P_0=1.00 GW, and that L=1000 km. The exercise can be solved without knowledge of R_0.



a) Assume ΔV=800 kV, a typical value for a long-distance HVDC system. Calculate the current in the cables.

b) Assuming the resistivity of the cables, ρ, to be known, set up expressions for the resistance of the cables as function of their diameter d (assuming a circular cross section), the voltage drop over each cable, and the total fraction of P_0 lost due to resistance in the cables.

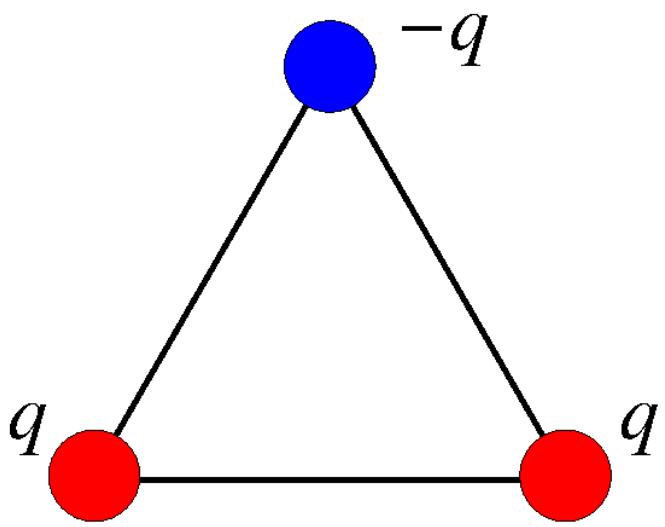
c) If the cables are made of Cu, what must d be in order to limit the loss to 5% of P_0? What will be the mass of these cables, without insulation etc.? Repeat the calculation for cables made of Al. The relevant constants can be found in Table 9.1 of the textbook, and in the previous exercise.

If time permits, find current market prices for Cu and Al, and estimate a lower limit for the price of these cables.

d) With the diameters found in c), calculate the loss due to resistive heating in percent of P_0 as a function of ΔV for ΔV ranging from 200 kV to 1000 kV.

Exercise 5.

Two positive point charges and one negative point charge are placed at the three corners of an equilateral triangle as shown in the figure. All charges have the same magnitude. The zero point of the electrostatic potential energy is set at infinity. Which of the following statements are correct?



A) The total potential energy of the three charges is positive

B) The total electrostatic force on the negative charge is directed vertically downward

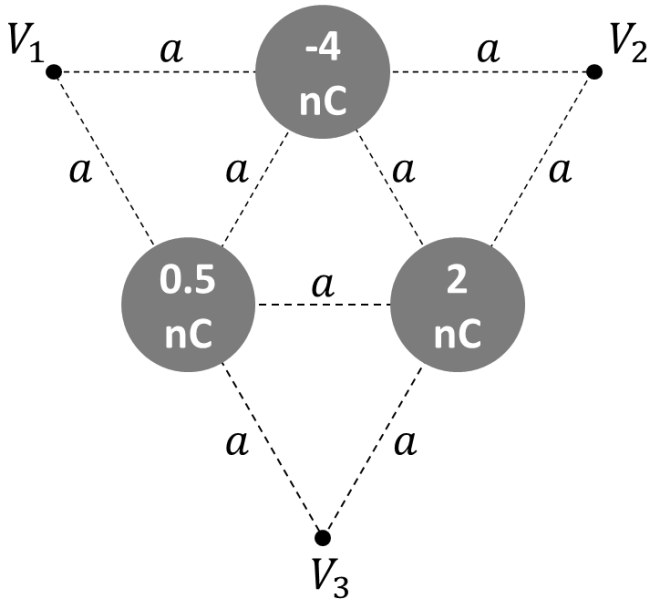
C) The total electrostatic force on the left positive charge is directed straight to the left

D) The electric field at the midpoint between the two positive charges is directed straight up

E) There are one or more zero points for the electrostatic potential within the sides of the triangle

F) Don't know

Exercise 6.



Three point charges whose values are -4 nC, 0.5 nC and 2 nC are distributed on an equilateral triangle with side length a as shown in the figure.

How are the values of the total electrical potential V_1, V_2 og V_3, as measured in the points shown on the figure, related?

A: $V_1 > V_2 > V_3$

B: $V_1 > V_3 > V_2$

C: $V_2 > V_1 > V_3$

D: $V_2 > V_3 > V_1$

E: $V_3 > V_1 > V_2$

F: $V_3 > V_2 > V_1$

G: Don't know.